

00_EXECUTIVE_OVERVIEW: Aura-Guard v1.3

Status: Confidential — Executive Summary
Version: 1.3 "Atom-Grade"
Subject: Deterministic AI Evidence Infrastructure

1. The Problem: The "Black Box" Liability

As organizations integrate Large Language Models (LLMs) into core business processes, they face a critical infrastructure gap: **AI systems are fundamentally non-auditable.**

- **Lack of Evidence:** Traditional logs are easily manipulated and cannot prove that a specific decision was made at a specific time.
- **Non-Reproducibility:** Probabilistic AI behavior makes it nearly impossible to replicate a past interaction for regulatory review.
- **Compliance Risk:** Under the **EU AI Act** and **DORA**, failure to provide a transparent, immutable audit trail can lead to severe legal and operational consequences.

2. The Solution: Aura-Guard v1.3

Aura-Guard is a lightweight, **Atom-Grade Deterministic Trust Infrastructure**. It acts as a cryptographic "black box recorder" for all AI interactions. Unlike standard logging systems, Aura-Guard ensures that every AI decision is **reproducible, cryptographically signed, and mathematically linked** to its input through an immutable hash chain.

3. Core Value Propositions

Feature	Executive Benefit
Cryptographic Evidence Chain	Provides non-repudiation. Every audit event is linked to the previous one, making the log tamper-proof.
Deterministic Replayability	Any past decision can be perfectly reproduced and verified by auditors using the Replay CLI tool.
Local-First & Air-Gapped	Zero data leakage. All processing remains on-premise, satisfying the strictest privacy requirements (GDPR/Banking).
Fail-Closed Security	The system prioritizes integrity; if a policy is tampered with, the system halts rather than providing an unverified audit.

4. Strategic Positioning

Aura-Guard moves your AI strategy from "Probabilistic Risk" to "Deterministic Governance." It is not just a tool; it is the **Evidence Fabric** that makes AI deployment defensible before regulators and stakeholders.

5. Compliance Readiness

The v1.3 architecture is specifically designed to meet the high-water mark of global regulations:

- **EU AI Act:** Articles 12-15 (Record-keeping, Transparency, and Human Oversight).
- **DORA:** Operational resilience and auditability for financial entities.
- **SOC2/ISO 27001:** Immutable logging and clear separation of duties.

"Every AI decision becomes reproducible and cryptographically verifiable."